

Multilevel Monte Carlo for Stochastic PDEs

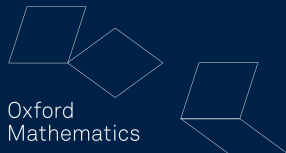


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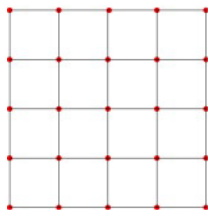
MMSC Viva Voce, Sep 2025



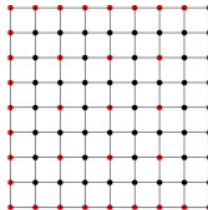
There are three main questions my dissertation aimed to answer:

1. What cost savings can MLMC achieve over standard MC for two representative SPDEs?
2. What are the effects of different coupling mechanisms?
3. What further performance improvements can parallelisation achieve?

Multilevel Monte Carlo Implementation



Coarse Mesh



Fine Mesh

Stochastic Heat Equation (SHE)

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + \xi$$

$$u(0, x) = 0, \quad u(t, 0) = u(t, 1) = 0$$

$$P = \int_0^1 u^2 dx \quad P = \left(\int_0^1 u(x, T) \sin(n\pi x) dx \right)^2$$

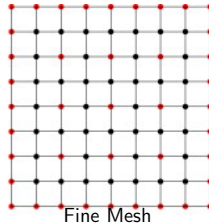
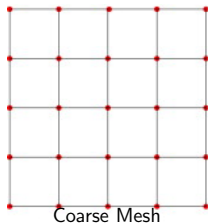
Dean-Kawasaki (DK) Equation

$$\frac{\partial \rho}{\partial t} = \frac{1}{2} \frac{\partial^2 \rho}{\partial x^2} + N_p^{-\frac{1}{2}} \frac{\partial(\sqrt{\rho} \xi)}{\partial x}$$

$$\rho(0, x) = \rho_0(x), \quad \text{periodic}$$

$$P = N_p \left(\int_0^{2\pi} (\rho - \bar{\rho}) \sin(x) dx \right)^2$$

Multilevel Monte Carlo Implementation



Stochastic Heat Equation (SHE)

$$U_j^{n+1} = U_j^n + \lambda(U_{j+1}^n - 2U_j^n + U_{j-1}^n) + \Delta W_j^n$$

$$U_j^0 = 0, \quad U_0^n = U_J^n = 0$$

$$\hat{P} = (U^n, U^n)_{\Delta x} \quad \hat{P} = (U, \phi_n)_{\Delta x}^2$$

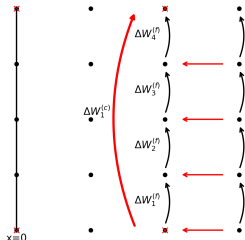
Dean-Kawasaki (DK) Equation

$$\rho_j^{n+1} = \rho_j^n + \frac{\lambda}{2} (\rho_{j+1}^n - 2\rho_j^n + \rho_{j-1}^n) + \frac{1}{\sqrt{N_p}} \frac{F_{j+1}^n - F_{j-1}^n}{2\Delta x}$$

$$\rho_j^0 = \rho_0(x_j), \quad \rho_0^n = \rho_J^n,$$

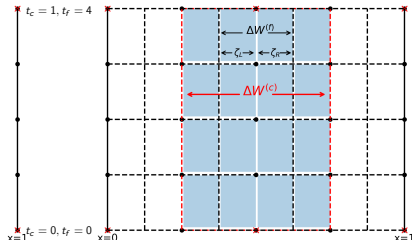
$$P = N_p ((\rho^n - \bar{\rho}^n, \sin(\Delta x))_{\Delta x})^2$$

Coupling mechanisms



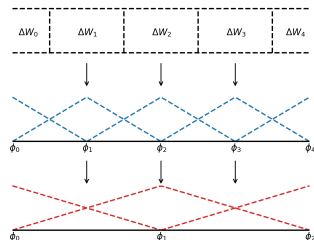
Nearest-Neighbour

- Used by Cornalba & Fischer.
- Coarse noise = sum of underlying + fine neighbour.



Central Coupling

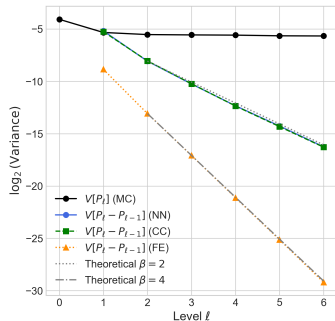
- Introduced in this dissertation.
- Mesh split into half-cells.
- Coarse noise built from shared half-cell increments.



Finite Element

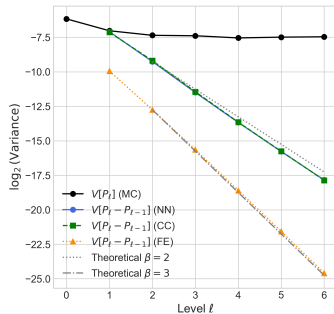
- Introduced in this dissertation.
- Noise projected onto hat basis.
- Coarse hats formed as sums of fine hats.

Effects of Coupling Mechanisms



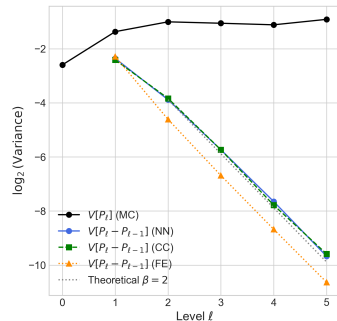
SHE - Squared Amplitude

- ▶ CC and NN coupling equivalent
- ▶ FE coupling achieves improved variance decay.



SHE - Energy

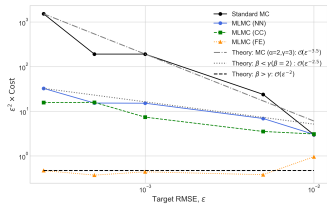
- ▶ CC and NN coupling equivalent.
- ▶ FE coupling achieves improved variance decay.



DK - Mean Fluctuations

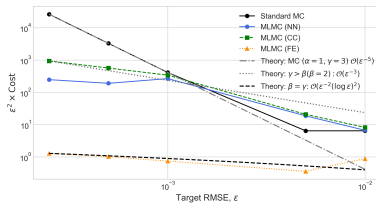
- ▶ All coupling mechanisms achieve equivalent variance decay.

Performance Results



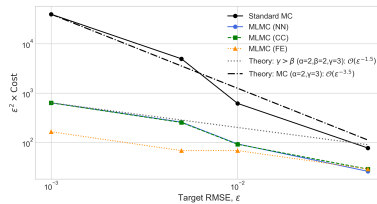
SHE - Squared Amplitude

- FE coupling: optimal scaling $\mathcal{O}(\epsilon^{-2})$.
- NN and CC: both suboptimal - $\mathcal{O}(\epsilon^{-2-(\gamma-\beta)/\alpha})$.



SHE - Energy

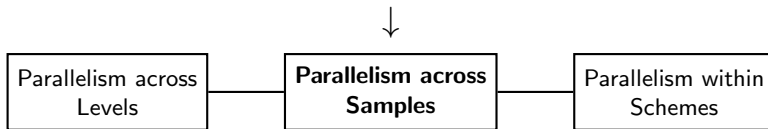
- FE coupling: near-optimal $\mathcal{O}(\epsilon^{-2}(\log \epsilon)^2)$.
- NN and CC: both suboptimal $\mathcal{O}(\epsilon^{-2-(\gamma-\beta)/\alpha})$.



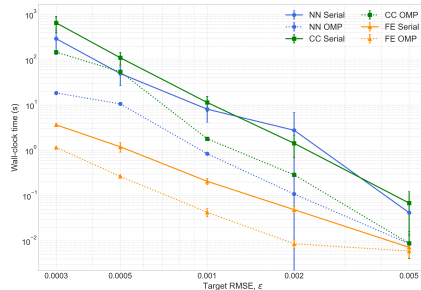
DK - Mean Fluctuations

- FE, NN and CC: equivalent scaling $\mathcal{O}(\epsilon^{-2-(\gamma-\beta)/\alpha})$.

Parallelism in MLMC



- ▶ Parallelised MLMC algorithm over samples using OpenMP.
- ▶ Resulted in average $\times 4$ speed improvement.



FE Coupling for DK equations

Let ϕ be the hat basis functions, and our approximate solution be

$$\rho(t, x) = \sum_{j=1}^{J-1} \rho_j(t) \phi_j(x).$$

$$\left(\frac{\partial \rho}{\partial t}, \phi_i \right)_{\Delta x} = \frac{1}{2} \left(\frac{\partial^2 \rho}{\partial t^2}, \phi_i \right)_{\Delta x} + N_p^{-1/2} \left(\frac{\partial \sqrt{\rho} \xi}{\partial x}, \phi_i \right)_{\Delta x}$$

$$M d\rho(t) + \frac{1}{2} K \rho(t) dt = d\eta$$

$$\int_0^{2\pi} \frac{\partial(\sqrt{\rho} \xi)}{\partial x} \phi_i dx = \int_0^{2\pi} \sqrt{\rho} \xi \phi'_i dx \quad \phi'_i = \begin{cases} \frac{1}{\Delta x}, & x \in [x_i - \Delta x, x_i] \\ -\frac{1}{\Delta x}, & x \in [x_i, x_i + \Delta x] \end{cases}$$